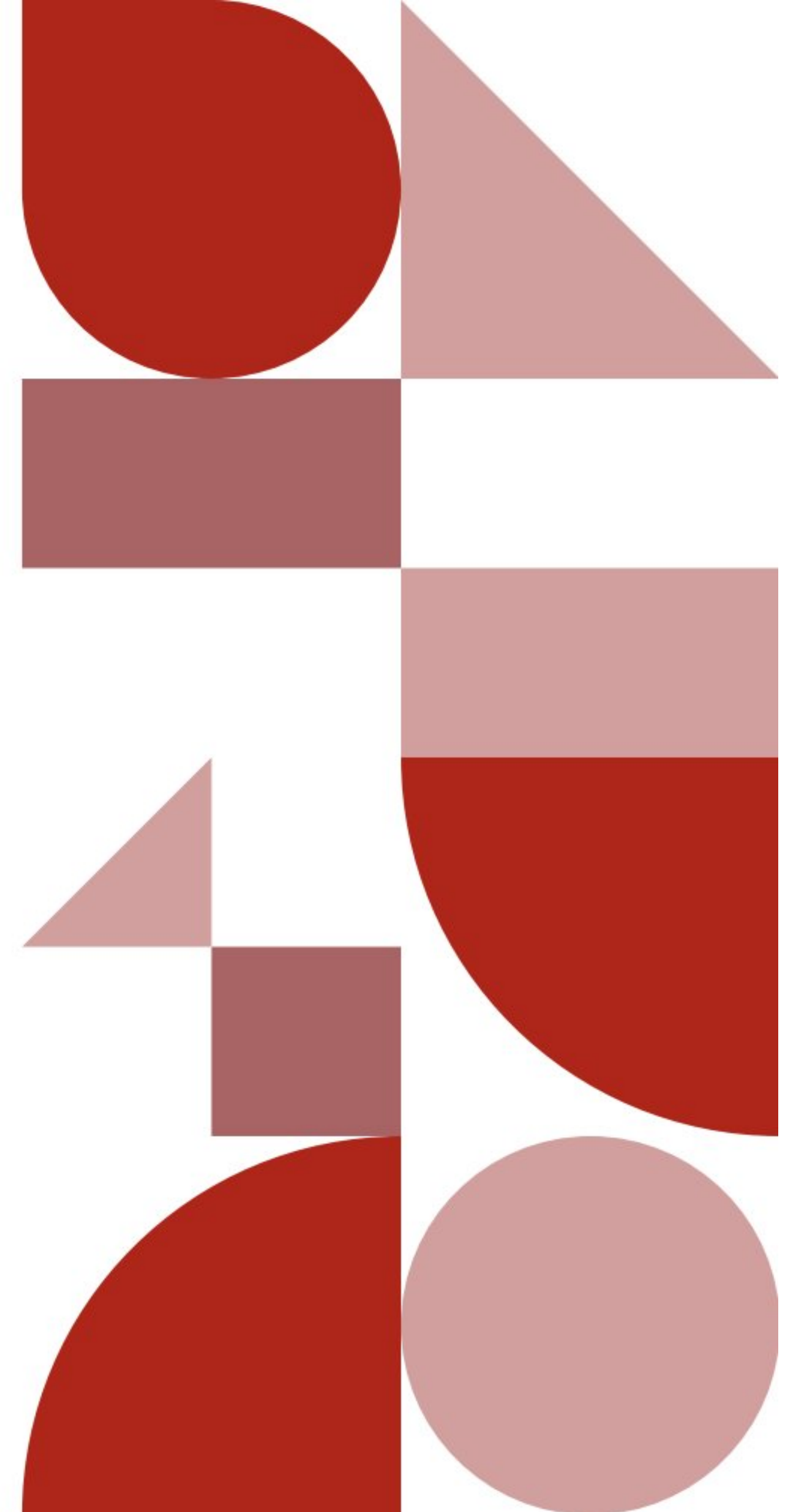




WEEK 8

CS50 INTRODUCTION INTO COMPUTER SCIENCE



JAVASCRIPT

Event Handling

JAVASCRIPT

1- addEventListener

```
<> index.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title></title>
5  </head>
6  <body>
7    <!-- <button id="myButton">hi</button> -->
8
```

```
JS script.js > ...
1  // 1-Adding Event Listeners Using addEventListener
2
3  Select the button
4  let button = document.querySelector("#myButton");
5
6  // Add a click event listener
7  button.addEventListener("click", () => {
8    console.log("Button clicked!");
9  });
10
11
12
13  You can attach multiple listeners to the same element and event.
14  button.addEventListener("click", () => console.log("First handler"));
15  button.addEventListener("click", () => console.log("Second handler"));
16
```

OUTPUT IN CONSOLE :

Button clicked!

First handler

Second handler

JAVASCRIPT

2- InlineEventHandler

```
9  
10 <!-- <button onclick="sayHello()">Click Me</button> -->  
11
```

```
16  
17 <!-- <script>  
18 function sayHello() {  
19     console.log("Hello, World!");  
20 }  
21 </script> -->  
22
```

OUTPUT IN CONSOLE :

Hello, World!

```
18  
19 // 2-Inline Event Handlers  
20 // You can also specify event handlers directly in HTML using attributes like onclick, onsubmit, etc.(code in html)  
21
```

3- Common Events

```
<> index.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title></title>
5  </head>
6  <body>
7    <!-- <button id="myButton">hi</button> -->
8
9
10   <!-- <button onclick="sayHello()">Click Me</button> -->
11
12   <form id="myForm">
13     <input id="textInput" type="text" placeholder="Enter text" />
14     <button type="submit">Submit</button>
15   </form>
```

OUTPUT IN CONSOLE :

Button clicked! (Click)

Key pressed: a (Pressing [a])

Form submitted! (Submit)

```
24  //3-Common Events
25  click
26  document.querySelector("#myButton").addEventListener("click", () => {
27    console.log("Button clicked!");
28  });
29
30
31  // //Keyup Event
32  document.querySelector("#textInput").addEventListener("keyup", (event) => {
33    console.log("Key pressed:", event.key);
34  });
35
36
37  //Submit Event
38  document.querySelector("form").addEventListener("submit", (event) => {
39    event.preventDefault(); // Prevent the page from reloading
40    console.log("Form submitted!");
41  });
42
```


JAVASCRIPT

Edx Problems

JAVASCRIPT

- **Example#1**

```
<> index.html > ...
1  <html lang="en">
2    <head>
3      <script>
4        function greet()
5        {
6          alert('hello, ' + document.querySelector('#name').value);
7        }
8      </script>
9      <title>hello</title>
10    </head>
11    <body>
12      <form onsubmit="greet(); return false;">
13        <input autocomplete="off" autofocus id="name" placeholder="Name" type="text">
14        <input type="submit">
15      </form>
16    </body>
17  </html>
```

OUTPUT IN WINDOW ALERT :

hello, ahmed

(if entering : ahmed)

JAVASCRIPT

- **Example#2**

```
23 <html lang="en">
24 <head>
25 <script>
26
27     document.addEventListener('DOMContentLoaded', function() {
28         document.querySelector('form').addEventListener('submit', function(e) {
29             alert('hello, ' + document.querySelector('#name').value);
30             e.preventDefault();
31         });
32     });
33 </script>
34 <title>hello</title>
35 </head>
36 <body>
37 <form>
38     <input autocomplete="off" autofocus id="name" placeholder="Name" type="text">
39     <input type="submit">
40 </form>
41 </body>
42 </html>
```

OUTPUT IN WINDOW ALERT :

hello, ahmed

(if entering : ahmed)

(but here wait for DOMContentLoaded

JAVASCRIPT

• Example#3

```
47 <html lang="en">
48   <head>
49     <title>background</title>
50   </head>
51   <body>
52     <button id="red">R</button>
53     <button id="green">G</button>
54     <button id="blue">B</button>
55     <script>
56
57       let body = document.querySelector('body');
58       document.querySelector('#red').addEventListener('click', function() {
59         body.style.backgroundColor = 'red';
60       });
61       document.querySelector('#green').addEventListener('click', function() {
62         body.style.backgroundColor = 'green';
63       });
64       document.querySelector('#blue').addEventListener('click', function() {
65         body.style.backgroundColor = 'blue';
66       });
67
68     </script>
69   </body>
70 </html>
```

OUTPUT IN CSS :

Red color for Body (Red Button)

Green color for Body (Green Button)

Blue color for Body (Blue Button)

JAVASCRIPT

• Example#4

```
76 <html lang="en">
77   <head>
78     <script>
79       // Toggles visibility of greeting
80       function blink()
81       {
82         let body = document.querySelector('body');
83         if (body.style.visibility == 'hidden')
84         {
85           body.style.visibility = 'visible';
86         }
87         else
88         {
89           body.style.visibility = 'hidden';
90         }
91       }
92     }
93
94     // Blink every 500ms
95     window.setInterval(blink, 500);
96
97   </script>
98   <title>blink</title>
99 </head>
100 <body>
101   hello, world
102 </body>
103 </html>
```

OUTPUT IN HTML :
(Blinking hello, world)

1- hello, world

2- " "

3- hello, world

4- " "

• Example#5

```
109 <html lang="en">
110
111 <head>
112
113   <title>autocomplete</title>
114 </head>
115
116 <body>
117
118   <input autocomplete="off" autofocus placeholder="Query" type="text">
119
120   <ul></ul>
121
122   <script src="large.js"></script>
123   <script>
124
125     let input = document.querySelector('input');
126     input.addEventListener('keyup', function(event) {
127       let html = '';
128       if (input.value) {
129         for (word of WORDS) {
130           if (word.startsWith(input.value)) {
131             html += `<li>${word}</li>` ;
132           }
133         }
134       }
135       document.querySelector('ul').innerHTML = html;
136     });
137
138   </script>
139
140 </body>
141 </html>
```

```
JS large.js > ...
1 // Define an array of words for the autocomplete
2 const WORDS = [
3   "apple",
4   "banana",
5   "cherry",
6   "date",
7   "elderberry",
8   "fig",
9   "grape",
10  "honeydew",
11  "kiwi",
12  "lemon",
13  "mango",
14  "nectarine",
15  "orange",
16  "papaya",
17  "quince",
18  "raspberry",
19  "strawberry",
20  "tangerine",
21  "ant",
22  "watermelon"
23 ];
```

JAVASCRIPT

- **Example#6**

```
148 <html lang="en">
149   <head>
150     <title>geolocation</title>
151   </head>
152   <body>
153     <script>
154       navigator.geolocation.getCurrentPosition(function(position) {
155         console.log(position.coords.latitude + ", " + position.coords.longitude);
156       });
157     </script>
158   </body>
159 </html>
```




Thank you!

